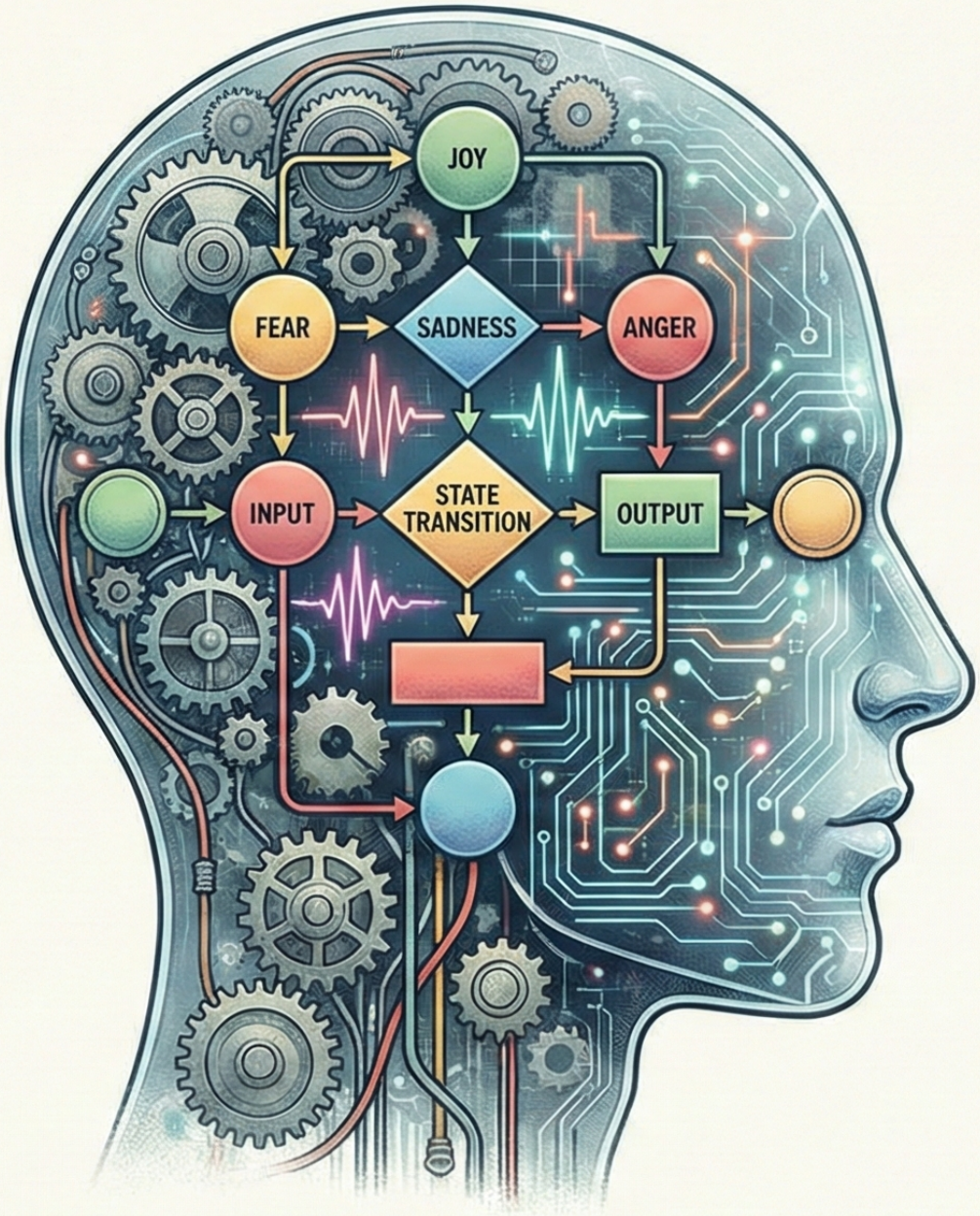


MY VIEW OF EMOTIONS THROUGH AUTOMATA THEORY

A Computational Perspective on Human Feeling

State of Emotions

and Automata Theory

Uttam Mahata

December 31, 2025

To the endless loops we call feelings,
and the halting problems we call love.

Preface

> *This is not a psychology book. It is not a textbook on automata theory either. It is something in between—a personal, intellectual exercise in understanding the chaos of human emotions through the lens of computation.*
>
> *I do not claim that the heart is literally a finite automaton, nor that love is an algorithm running on biological hardware. What I do claim is that the formalisms of theoretical computer science offer a surprisingly rich vocabulary for articulating what it* feels like *to experience emotions: the transitions, the states, the inputs that push us from one configuration to another.*
>
> *The models here are* ***metaphors with mathematical discipline****. They are not psychological claims waiting for empirical validation. They are tools for thought—ways of seeing that I have found personally illuminating.*

Perhaps you picked up this book because you love automata theory and want to see it applied unconventionally. Perhaps you are seeking a different way to think about your own emotional life. Or perhaps you are simply curious about what happens when formal systems meet the messiness of being human.

Whatever brought you here, I invite you to read with an open mind. The proofs are real. The feelings are real. The connection between them is mine to offer and yours to interpret.

—Uttam, December 2025

Contents

Preface **iii**

I Foundations **1**

1 Why Model Emotions Mathematically? **2**
1.1 The Question . . . 2
1.2 Why Automata? . . . 2
1.3 The Limits of the Metaphor . . . 3

2 The Emotional Automaton **4**
2.1 Defining the Emotional State Space . . . 4
2.2 A Basic State Diagram . . . 4
2.3 The Transition Function . . . 5
2.4 Accepting States: What Does It Mean to "Accept"? . . . 5
2.5 Non-Determinism and Probability . . . 6

II Love as Computation **7**

3 Love as a Transient State **8**
3.1 Love Is Not Special . . . 8
3.2 The Love Subautomaton . . . 8
3.3 Why Love Is "Overhyped" . . . 9
3.4 The Self-Loop of Sustained Love . . . 9

4 The Turing Machine of Cognition **11**
4.1 Beyond Finite Automata . . . 11
4.2 The Tape as Memory . . . 12

5 The Undecidability of "True Love" **13**
5.1 The Decision Problem . . . 13
5.2 Reduction from the Halting Problem . . . 13
5.3 Interpretation . . . 14

6 Rice's Theorem and the Properties of Love **15**
6.1 Rice's Theorem . . . 15
6.2 Application to Emotional Duration . . . 15

III Other Emotional States **16**

7 The Automaton of Grief **17**
7.1 Grief as a State Space . . . 17
7.2 Non-Deterministic Grief . . . 17
7.3 Acceptance as an Absorbing State? . . . 18

8 The Automaton of Joy **19**
8.1 Joy as a Transient State . . . 19
8.2 The Fragility of Joy . . . 19

9 The Automaton of Anger **21**
9.1 Anger as a High-Energy State . . . 21
9.2 The Danger of Rage . . . 21

10 The Automaton of Fear **22**
10.1 Fear as a Protective State . . . 22
10.2 Anxiety as a Waiting State . . . 22

IV Deeper Incompleteness **24**

11 Gödel and Emotional Self-Knowledge **25**
11.1 The Limits of Introspection . . . 25
11.2 What We Cannot Know About Ourselves . . . 25
11.3 The Need for External Perspective . . . 26

12 Self-Reference and Emotional Paradox **27**
12.1 The Liar's Paradox of Feeling . . . 27
12.2 Recursive Emotions . . . 27

13 Fixed Points of Feeling **28**
13.1 The Fixed-Point Theorem . . . 28
13.2 Emotional Fixed Points . . . 28
13.3 Breaking Fixed Points . . . 28

14 The Limits of Emotional Computation **29**
14.1 What We Have Learned . . . 29
14.2 What the Model Cannot Capture . . . 29
14.3 Final Reflection . . . 30

Formal Definitions **31**
.1 Finite Automata . . . 31
.2 Turing Machines . . . 31
.3 The Halting Problem . . . 32
.4 Rice's Theorem . . . 32

Notation Reference **33**

Part I

Foundations

Chapter 1

Why Model Emotions Mathematically?

> "The heart has its reasons, which reason does not know."
>
> —Blaise Pascal

1.1 The Question

> *I remember the first time I tried to explain to a friend why I was sad. The words came out wrong. They always do. "I feel... heavy," I said. "Like something is pulling me down, but I don't know what." My friend nodded, but I could see they didn't quite understand. How could they? I didn't understand it myself.*

Emotions resist articulation. They are slippery, contradictory, and intensely private. We gesture at them with metaphors—we "fall" in love, "burst" with joy, "sink" into depression—but the metaphors never quite capture the reality.

This book is an experiment in finding a new vocabulary. Specifically, I want to borrow the language of **automata theory**: the study of abstract machines, their states, and the inputs that drive transitions between them.

1.2 Why Automata?

Automata theory provides:

1. **A finite set of states** — We are not infinite beings. At any moment, we occupy some configuration of feeling.

2. **Inputs** — External stimuli trigger changes. A word, a touch, a memory.

3. **Transitions** — The rules (however complex) that govern how we move from state to state.

4. **Acceptance and rejection** — Some states feel like destinations; others like way stations.

FORMAL INTERLUDE.

An automaton is defined as a 5-tuple:

$$M = (Q, \Sigma, \delta, q_0, F)$$

where Q is a finite set of states, Σ is an input alphabet, $\delta : Q \times \Sigma \to Q$ is the transition function, $q_0 \in Q$ is the initial state, and $F \subseteq Q$ is the set of accepting states.

I am not claiming the brain is literally a finite automaton. The brain has roughly 86 billion neurons forming 100 trillion connections—far beyond any finite model. But when I introspect on my emotional life, it feels *finite. There are only so many flavors of feeling. Sadness, joy, anger, fear, love, indifference... the categories blur at the edges, but they are countable.*

1.3 The Limits of the Metaphor

Every metaphor has boundaries. Automata theory cannot tell you *why* you feel the way you do. It cannot offer therapy or cure. What it offers is **structure**—a way to map the terrain, even if the map is not the territory.

In the chapters that follow, we will explore:

- How emotional states can be modeled as automaton configurations
- How different emotions (love, grief, joy, anger, fear) have distinct "transition signatures"
- How the concept of decidability applies to questions like "Will this love last?"
- How Gödel's incompleteness theorems illuminate the limits of emotional self-knowledge

I do not expect you to agree with everything here. I only hope that, by the end, you will have a new way of thinking about the states you pass through every day.

Chapter 2

The Emotional Automaton

> "We are what we repeatedly do."
>
> —Will Durant (paraphrasing Aristotle)

2.1 Defining the Emotional State Space

Let us begin with the foundational abstraction: the **emotional automaton**.

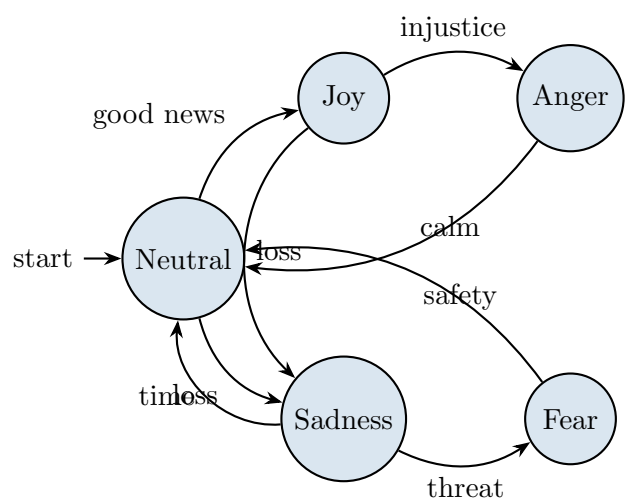
Definition 2.1 (Emotional Automaton)**.** An emotional automaton is a 5-tuple $\mathcal{E} = (Q, \Sigma, \delta, q_0, F)$ where:

- Q is a finite set of emotional states (e.g., {Neutral, Joy, Sadness, Anger, Fear, Love})
- Σ is a set of external stimuli (e.g., {Attention, Neglect, Conflict, Validation, Loss})
- $\delta : Q \times \Sigma \to Q$ is the transition function
- $q_0 \in Q$ is the initial state (typically Neutral)
- $F \subseteq Q$ is the set of "stable" or "accepting" states

> *When I wake up in the morning, I am usually in something like a Neutral state. Then the day happens. An email arrives—Conflict. A friend calls—Validation. Each input nudges me through the state space until, by evening, I find myself somewhere I did not plan to be.*

2.2 A Basic State Diagram

Consider a simple emotional automaton with five states:



Remark. Notice that not all states are equally "reachable" from all other states. Sadness, for instance, can be reached from Joy (via loss) but Joy cannot be reached directly from Fear without first returning to Neutral.

2.3 The Transition Function

The heart of the automaton is δ—the transition function. In formal terms:

FORMAL INTERLUDE.

$$\delta : Q \times \Sigma \to Q$$

Given a current state q and an input symbol σ, the function returns the next state.

Examples:

$$\begin{aligned} \delta(\text{Neutral}, \text{good news}) &= \text{Joy} \\ \delta(\text{Joy}, \text{loss}) &= \text{Sadness} \\ \delta(\text{Anger}, \text{calm}) &= \text{Neutral} \end{aligned}$$

Of course, real emotional transitions are not deterministic. The same input can produce different outputs depending on context, history, mood. But the deterministic model is a starting point—a skeleton on which we can later add probabilistic flesh.

2.4 Accepting States: What Does It Mean to "Accept"?

In classical automata theory, a string is accepted if the automaton ends in an accepting state $q \in F$ after processing all inputs. But what does "acceptance" mean emotionally?

I propose two interpretations:

1. **Stability**: An accepting state is one where we can rest. We are not immediately pushed elsewhere.

2. **Satisfaction**: An accepting state represents some form of equilibrium or fulfillment.

   In my own life, "acceptance" rarely means permanent residence. It means temporary shelter—a state where I can pause before the next input arrives and the next transition begins.

2.5 Non-Determinism and Probability

Real emotions exhibit **non-determinism**. The same stimulus can provoke different responses on different days. To capture this, we extend to a **probabilistic** or **non-deterministic finite automaton (NFA)**.

Definition 2.2 (Probabilistic Emotional Automaton)**.** A probabilistic emotional automaton extends the transition function:

$$\delta : Q \times \Sigma \to \mathcal{P}(Q) \quad \text{or} \quad \delta : Q \times \Sigma \times Q \to [0, 1]$$

where the transition includes probability weights.

Example 2.1. Suppose you are in the Neutral state and receive criticism (σ = Conflict). There is:

- 40% chance you transition to Sadness
- 35% chance you transition to Anger
- 25% chance you remain Neutral (thick skin)

This probabilistic view aligns better with lived experience. We are not machines that respond identically every time. We are distributions over possible responses.

Part II

Love as Computation

Chapter 3

Love as a Transient State

> "The course of true love never did run smooth."
>
> —William Shakespeare, *A Midsummer Night's Dream*

3.1 Love Is Not Special

Society tells us that love is the ultimate destination. Songs, films, novels—they all orbit around the moment of falling in love, as if reaching that state is the purpose of existence. But when I examine my own experience, I find that love is just... a state. One among many. It comes. It goes. It returns. It leaves again.

In our automaton framework, let us formalize this observation.

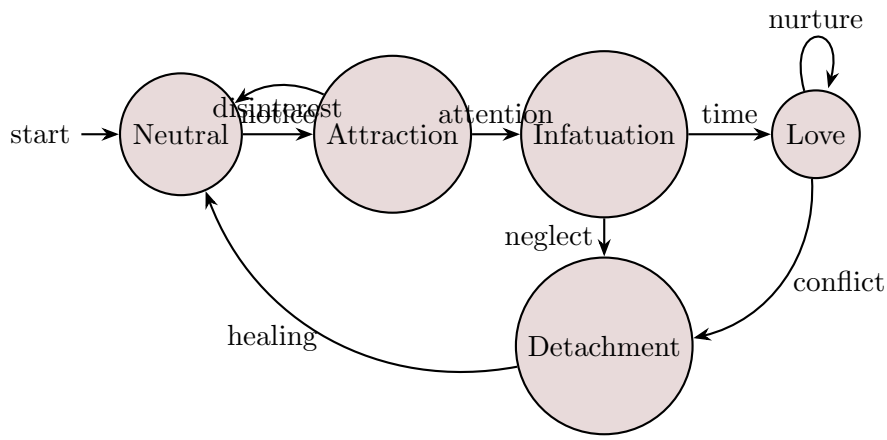
Definition 3.1 (Love as a State)**.** Let $L \in Q$ denote the state of Love. We claim:

1. L is not necessarily in F (the set of accepting states)
2. L is not an "absorbing" state (there exist transitions out of L)
3. The transition probabilities into and out of L are non-zero

3.2 The Love Subautomaton

Consider a more refined state space focused on romantic attachment:

$$Q_{\text{love}} = \{\text{Neutral}, \text{Attraction}, \text{Infatuation}, \text{Love}, \text{Detachment}\}$$



Formal Interlude.

The transition function for this subautomaton:

$$\begin{aligned}
\delta(\text{Neutral}, \text{notice}) &= \text{Attraction} \\
\delta(\text{Attraction}, \text{attention}) &= \text{Infatuation} \\
\delta(\text{Infatuation}, \text{time}) &= \text{Love} \\
\delta(\text{Love}, \text{neglect}) &= \text{Detachment} \\
\delta(\text{Love}, \text{nurture}) &= \text{Love} \quad \text{(self-loop)}
\end{aligned}$$

3.3 Why Love Is "Overhyped"

From a computational perspective, the Love state has no special properties:

1. **No additional computational power**: The automaton does not become more expressive upon entering L.
2. **Not necessarily accepting**: If F represents stable fulfillment, $L \notin F$ for many individuals. Love is often anxious, uncertain, demanding.
3. **High transition probability out**: Statistically, most love relationships end—through breakup, death, or gradual fade.

I do not say this to be cynical. I say it because recognizing love as "just another state" frees us from the pressure of treating it as salvation. Love is beautiful. But so is joy. So is peace. The emotional automaton has many worthy destinations.

3.4 The Self-Loop of Sustained Love

The only way love persists is through **continuous input**. The self-loop $\delta(L, \text{nurture}) = L$ requires constant feeding:

- Attention

- Effort
- Vulnerability
- Repair after conflict

Without these inputs, the automaton transitions elsewhere. Love is not a place you arrive at and stay. It is a place you must keep choosing.

Chapter 4

The Turing Machine of Cognition

> "We can only see a short distance ahead, but we can see plenty there that needs to be done."
>
> —Alan Turing

4.1 Beyond Finite Automata

Finite automata are elegant but limited. They have no memory beyond their current state. But emotions involve **memory**:

- We remember past hurts
- We anticipate future losses
- We ruminate, replay, revise

To model this, we need a more powerful formalism: the **Turing Machine**.

Definition 4.1 (Emotional Turing Machine)**.** An emotional Turing machine is defined as:

$$M = (Q, \Sigma, \Gamma, \delta, q_0, q_{\text{accept}}, q_{\text{reject}})$$

where:

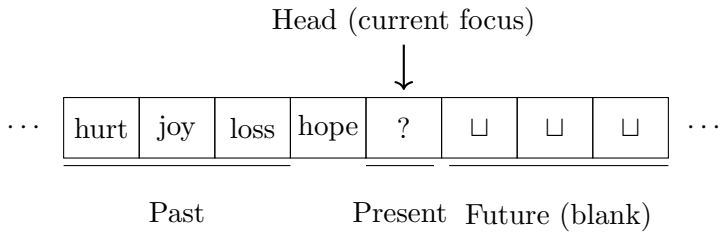
- Q = emotional-cognitive states
- Σ = external stimuli (input alphabet)
- Γ = internal memory contents (tape alphabet): beliefs, narratives, expectations
- $\delta : Q \times \Gamma \to Q \times \Gamma \times \{L, R, S\}$ = transition function
- The tape represents our infinite capacity for memory and rumination

  The tape is where I store everything: the time she didn't text back, the moment he said those words, the look on their face. The head moves left and right, revisiting old symbols, overwriting them with new interpretations. This is rumination. This is memory. This is what makes emotions so much more than simple stimulus-response.

4.2 The Tape as Memory

Consider the tape as containing:

- **Past experiences**: encoded as symbols
- **Current interpretations**: the head reads and writes
- **Future projections**: blank tape extending infinitely to the right



The Turing Machine can:

- Move left to revisit past experiences
- Overwrite memories with new interpretations
- Move right into unwritten future
- Loop indefinitely (rumination spirals)

Chapter 5

The Undecidability of "True Love"

> "The eternal mystery of the world is its comprehensibility."
>
> —Albert Einstein

5.1 The Decision Problem

We now arrive at a central question:

> **Given complete information about two people, can we determine whether their love will last forever?**

Let us formalize this as a decision problem.

Definition 5.1 (TRUE_LOVE Problem)**.**

$$\text{TRUE_LOVE} = \{\langle M \rangle \mid M \text{ is an emotional TM that enters state } L \text{ and never leaves}\}$$

In other words: is "forever love" decidable?

5.2 Reduction from the Halting Problem

Theorem 5.1. *TRUE_LOVE is undecidable.*

Proof. We reduce from the Halting Problem.

Let $\text{HALT} = \{\langle M, w \rangle \mid M \text{ halts on input } w\}$.

Given an instance $\langle M, w \rangle$ of HALT, construct an emotional Turing Machine M' as follows:

1. M' simulates M on w
2. If M halts, M' enters state L (Love) and remains there
3. If M does not halt, M' oscillates among emotional states forever

Then:

$$M \text{ halts on } w \iff M' \text{ enters } L \text{ and stays forever}$$

Thus:

$$\text{HALT} \leq_m \text{TRUE_LOVE}$$

Since HALT is undecidable, TRUE_LOVE is undecidable. □ □

5.3 Interpretation

> *There is no algorithm that can tell you whether your love will last. No amount of analysis, no compatibility test, no astrological chart, no psychological profile can decide this question in finite time. The very nature of the question places it beyond computation.*

This result has practical consequences:

Claim	**Computability**
"I love you now"	Decidable
"I loved you then"	Decidable
"I will love you forever"	**Undecidable**
"We are meant to be"	**Uncomputable**

The past and present are accessible. The eternal future is not.

Chapter 6

Rice's Theorem and the Properties of Love

6.1 Rice's Theorem

Theorem 6.1 (Rice's Theorem)**.** *Any non-trivial semantic property of a Turing machine is undecidable.*

A property is **non-trivial** if it is true for some TMs and false for others. A property is **semantic** if it concerns the behavior (rather than the syntax) of the machine.

6.2 Application to Emotional Duration

Consider the property:

> "Will this emotional state persist for duration t?"

This is a semantic property of the emotional Turing machine. It concerns behavior over time, not the structure of the transition rules.

Corollary 6.2. *The duration of Love (or any emotional state) is undecidable.*

Proof. "Duration $\geq t$" is a non-trivial semantic property. By Rice's Theorem, it is undecidable. □ □

> *We cannot compute how long love will last. We cannot compute whether grief will end. We cannot compute when joy will return. These are not failures of our methods—they are fundamental limits of computation itself.*

Part III

Other Emotional States

Chapter 7

The Automaton of Grief

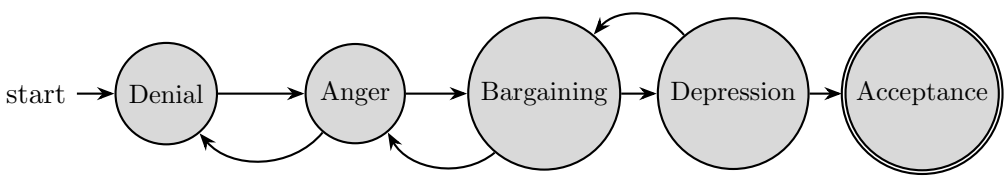
"Grief is the price we pay for love."

—Queen Elizabeth II

7.1 Grief as a State Space

Grief is not a single state but a **subautomaton** with its own internal structure. The famous Kübler-Ross model suggests five stages:

$$Q_{\text{grief}} = \{\text{Denial}, \text{Anger}, \text{Bargaining}, \text{Depression}, \text{Acceptance}\}$$



I have walked this automaton. Not in a straight line—no one does. You think you have reached Acceptance, and then a memory arrives as input, and suddenly you are back in Anger, or deeper in Depression. Grief is non-linear. The automaton has loops.

7.2 Non-Deterministic Grief

The transitions in grief are highly non-deterministic:

- A song can send you from Acceptance back to Depression
- A friend's words can push you from Anger to Bargaining
- Time alone may leave you stuck in Denial

Definition 7.1 (Grief NFA)**.** Grief is better modeled as a non-deterministic finite automaton where:

$$\delta : Q_{\text{grief}} \times \Sigma_{\text{grief}} \to \mathcal{P}(Q_{\text{grief}})$$

Each input can lead to multiple possible states.

7.3 Acceptance as an Absorbing State?

Is Acceptance in F? Is it absorbing (no transitions out)?

> *I don't think so. Acceptance is not the end of grief. It is learning to carry grief while moving forward. The automaton doesn't halt at Acceptance—it simply allows other states to coexist. You can be in Acceptance and still, on certain days, revisit Sadness. The automaton runs forever; it just learns to include more loops that pass through peace.*

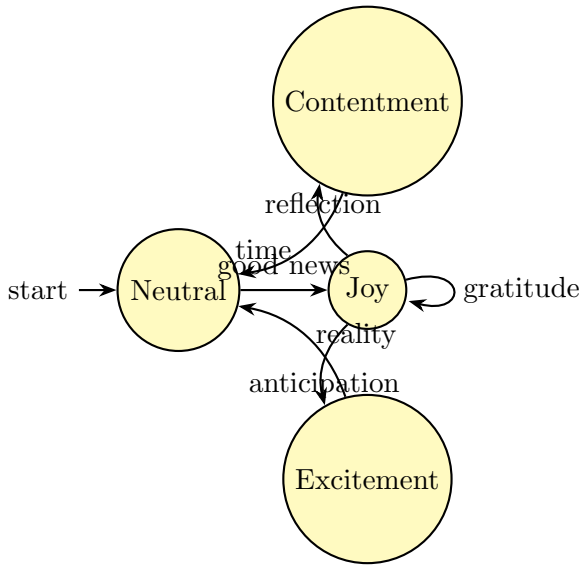
Chapter 8

The Automaton of Joy

> "Joy is the simplest form of gratitude."
>
> —Karl Barth

8.1 Joy as a Transient State

Joy, like love, is often treated as a destination. But in our model, Joy $\in Q$ is transient—easily entered, easily exited.



8.2 The Fragility of Joy

Joy has high transition probability out. Small inputs can disrupt it:

- A stray thought
- An unwelcome memory
- The anticipation of loss (even while happy)

Formal Interlude.

The expected residence time in Joy:

$$\mathbb{E}[\tau_J] = \frac{1}{\sum_\sigma P(\text{leave } J \mid \sigma)}$$

is typically short compared to other states.

I have learned not to grasp at joy. The tighter I hold it, the faster it slips away. Instead, I try to notice when I am in the Joy state—to observe it without clinging. The self-loop requires gratitude, not grasping.

Chapter 9

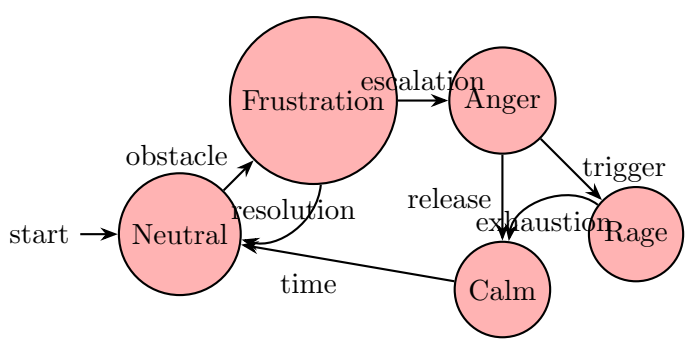
The Automaton of Anger

> "Anger is an acid that can do more harm to the vessel in which it is stored than to anything on which it is poured."
>
> —Mark Twain

9.1 Anger as a High-Energy State

Anger is characterized by:

- Rapid entry (low activation threshold)
- High energy expenditure
- Strong transition pressure (the state "wants" to act)



9.2 The Danger of Rage

Rage is a near-absorbing state with destructive outputs. The transition to Calm from Rage typically requires **exhaustion**—the energy depletes before the state can change.

> *I have been in Rage. It feels infinite in the moment—like there is no exit. But the automaton always eventually transitions. The question is what damage occurs before the transition completes.*

Chapter 10

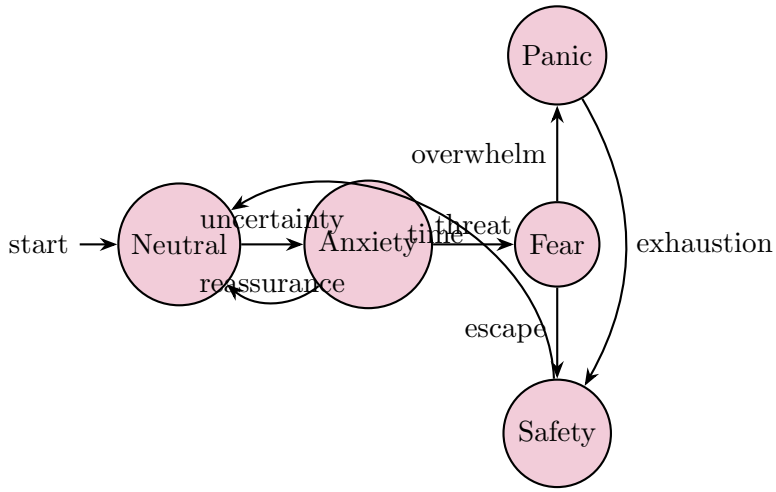
The Automaton of Fear

> "The only thing we have to fear is fear itself."
>
> —Franklin D. Roosevelt

10.1 Fear as a Protective State

Fear evolved as a survival mechanism. In automaton terms, it is a state that:

- Activates rapidly (fight-or-flight)
- Narrows the input alphabet (tunnel vision)
- Biases transitions toward safety-seeking



10.2 Anxiety as a Waiting State

Anxiety is the state of anticipating fear without a concrete threat. It is characterized by:

- Scanning for threatening inputs

- Low threshold for transition to Fear
- Difficulty returning to Neutral without external reassurance

FORMAL INTERLUDE.

Anxiety can be modeled as a state with:

$$P(\text{transition to Fear} \mid \text{any ambiguous input}) > P(\text{remain in Anxiety})$$

Anxiety is exhausting precisely because it is a waiting state. The automaton is poised for transition, consuming energy in readiness, but the threatening input may never arrive. The computation spins without progress.

Part IV

Deeper Incompleteness

Chapter 11

Gödel and Emotional Self-Knowledge

> "I don't know what I may seem to the world, but, as to myself, I seem to have been only like a boy playing on the sea-shore."
>
> —Isaac Newton

11.1 The Limits of Introspection

Gödel's First Incompleteness Theorem states:

Theorem 11.1 (Gödel's First Incompleteness Theorem)**.** *Any consistent formal system F capable of expressing basic arithmetic contains true statements that cannot be proven within F.*

Applied to emotional cognition:

> **Any sufficiently complex emotional system contains truths about itself that it cannot verify through introspection.**

11.2 What We Cannot Know About Ourselves

If our emotional cognition is modeled as a formal system, there exist:

- Emotions we feel but cannot articulate
- Patterns we enact but cannot recognize
- Motivations that drive us but remain invisible to introspection

> *I have spent years trying to understand why I react certain ways. Therapy, journaling, meditation—each reveals something, but never everything. The system cannot fully model itself from within. There is always a residue of mystery.*

11.3 The Need for External Perspective

Just as Gödel's theorem implies that truths about a system can sometimes only be seen from *outside* the system, emotional self-knowledge often requires external perspective:

- Friends who notice patterns we miss
- Therapists who name what we cannot
- Time, which reveals what the present obscures

FORMAL INTERLUDE.

Let $\mathcal{E}$ be the formal system of emotional self-knowledge. Then:

$$\exists \phi \text{ (emotional truth)} : \mathcal{E} \nvdash \phi \wedge \mathcal{E} \nvdash \neg\phi$$

Some emotional truths are undecidable from within.

Chapter 12

Self-Reference and Emotional Paradox

12.1 The Liar's Paradox of Feeling

Consider the statement:

> "I am not feeling anything right now."

If this statement is true, then you are feeling nothing—including the feeling that you feel nothing. But noticing that you feel nothing is itself a kind of feeling (awareness, blankness, perhaps subtle discomfort).

This is a self-referential paradox analogous to the liar's paradox in logic.

12.2 Recursive Emotions

Emotions can be recursive:

- Fear of fear (anxiety about becoming anxious)
- Guilt about anger (feeling bad for feeling bad)
- Joy at sadness ending (meta-relief)

Formal Interlude.

Let $E(x)$ denote "feeling emotion x." Then recursive emotions are:

$$E(E(x)) = \text{“feeling about feeling”}$$

These can nest arbitrarily: $E(E(E(x)))$, etc.

I once spent an entire day feeling anxious about the fact that I was feeling anxious. The meta-level consumed the base level. It was recursive descent without a base case—an infinite loop that only broke when exhaustion forced a halt.

Chapter 13

Fixed Points of Feeling

13.1 The Fixed-Point Theorem

In computability theory, the **Kleene Fixed-Point Theorem** states:

Theorem 13.1 (Kleene Fixed-Point Theorem)**.** *For any computable function f, there exists a program p such that p and $f(p)$ compute the same function.*

In simpler terms: for any transformation, there exists something that maps to itself.

13.2 Emotional Fixed Points

Applied to emotions, a **fixed point** is an emotional state that, when processed through our cognitive-emotional machinery, returns to itself:

$$\delta(q, \sigma_{\text{self}}) = q$$

Depression can be a fixed point. The thought "I am worthless" leads to behaviors that confirm "I am worthless," which reinforces the thought. The cycle is stable—not in a healthy sense, but in a mathematical sense. It resists perturbation.

13.3 Breaking Fixed Points

Therapy often aims to **destabilize fixed points**:

- Introduce new inputs that the self-loop doesn't expect
- Change the transition function through cognitive restructuring
- Add new states (perspectives) that provide escape routes

Formal Interlude.

To break a fixed point at state q:

1. Introduce $\sigma_{\text{new}} \notin \Sigma$ such that $\delta(q, \sigma_{\text{new}}) = q' \neq q$
2. Or modify δ to δ' where $\delta'(q, \sigma_{\text{self}}) \neq q$

Chapter 14

The Limits of Emotional Computation

> "There are more things in heaven and earth, Horatio, than are dreamt of in your philosophy."
>
> —William Shakespeare, *Hamlet*

14.1 What We Have Learned

Across this book, we have established:

1. **Emotions can be modeled as automata states**, with stimuli as inputs and behaviors as outputs.
2. **No emotional state is special**—not love, not joy, not even acceptance. They are all transient configurations.
3. **Complex emotions require Turing-complete models** with memory (the tape of experience).
4. **Key questions are undecidable**: Will love last? When will grief end? These lie beyond computation.
5. **Self-knowledge is incomplete**: Gödel's theorem applies to introspection.
6. **Emotions can be recursive and self-referential**, leading to paradoxes and fixed points.

14.2 What the Model Cannot Capture

I have spent this book building formal structures around feelings. But I am under no illusion that the structures are *the feelings. The map is not the territory. The automaton is not the heart.*

There is something about the texture of sadness—its weight, its color, its taste—that no transition function can encode. There is something about the flash of joy—its brightness, its brevity—that resists formalization.

These models are tools for thinking, not replacements for feeling.

14.3 Final Reflection

I began this project because I wanted to understand myself. I end it knowing that complete understanding is impossible—formally, provably impossible. And perhaps that is the most important lesson.

We are not puzzles to be solved. We are computations that run without halting, processing inputs we did not choose, transitioning through states we cannot predict, carrying a tape of memory that stretches back further than we can read.

The emotional automaton computes. It does not conclude. And maybe that is enough.

Formal Definitions

.1 Finite Automata

Definition .1 (Deterministic Finite Automaton (DFA))**.** A DFA is a 5-tuple $M = (Q, \Sigma, \delta, q_0, F)$ where:

- Q is a finite set of states
- Σ is a finite input alphabet
- $\delta : Q \times \Sigma \to Q$ is the transition function
- $q_0 \in Q$ is the initial state
- $F \subseteq Q$ is the set of accepting states

Definition .2 (Non-deterministic Finite Automaton (NFA))**.** An NFA is a 5-tuple $M = (Q, \Sigma, \delta, q_0, F)$ where the transition function is:

$$\delta : Q \times (\Sigma \cup \{\epsilon\}) \to \mathcal{P}(Q)$$

.2 Turing Machines

Definition .3 (Turing Machine)**.** A Turing machine is a 7-tuple $M = (Q, \Sigma, \Gamma, \delta, q_0, q_{\text{accept}}, q_{\text{reject}})$ where:

- Q is a finite set of states
- Σ is the input alphabet (not including blank symbol)
- Γ is the tape alphabet ($\Sigma \subseteq \Gamma$, includes blank $\sqcup$)
- $\delta : Q \times \Gamma \to Q \times \Gamma \times \{L, R\}$ is the transition function
- $q_0 \in Q$ is the initial state
- $q_{\text{accept}} \in Q$ is the accepting state
- $q_{\text{reject}} \in Q$ is the rejecting state

.3 The Halting Problem

Definition .4 (Halting Problem)**.**

$$\text{HALT} = \{\langle M, w\rangle \mid M \text{ is a TM that halts on input } w\}$$

Theorem .1. *HALT is undecidable.*

.4 Rice's Theorem

Theorem .2 (Rice's Theorem)**.** *Let P be any non-trivial property of partial computable functions. Then the set:*

$$\{\langle M\rangle \mid M \textit{ computes a function with property } P\}$$

is undecidable.

Notation Reference

Symbol	Meaning
Q	Set of states
Σ	Input alphabet (stimuli)
Γ	Tape alphabet (memory contents)
δ	Transition function
q_0	Initial state
F	Set of accepting (stable) states
L	Love state
$\sqcup$	Blank symbol (unwritten future)
$\mathcal{P}(X)$	Power set of X
$\leq_m$	Many-one reducibility